Smart India Hackathon 2026: Project Guide in Simple English

Project Details at a Glance

- **Problem Statement ID:** SIH26011
- **Problem Title:** 3D ULPIN Generation and Vertical Property Mapping System
- **Ministry / Organization:** Ministry of Rural Development (Government of India)
- **Category:** Software

1. The Problem Statement & Simple Explanation

What is ULPIN?

In India, every piece of land is given a unique 14-character code called **ULPIN** (Unique Land Parcel Identification Number), also known as **Bhu-Aadhaar**. Just like an Aadhaar card identifies a person, ULPIN identifies a piece of land based on its GPS coordinates.

What is the Real Problem?

ULPIN currently works **only in 2D (flat ground)**.

1. **The High-Rise Reality:** In growing Indian cities like Nashik, Pune, and Mumbai, land is limited. Builders construct 10 to 20-story apartment buildings on a single plot of land.
2. **One Plot, Many Owners:** While there is only one piece of ground at the bottom, there are 50 to 100 different families living vertically above it.
3. **Flats Have No Digital 3D Identity:** On official government digital maps, a flat on the 4th floor (Flat 402) has **no unique 3D geographical ID**. The map only shows the flat ground below.

Why Does This Cause Huge Trouble in Daily Life?

- **Bank Loan Scams (Double Mortgages):** Because there is no central 3D digital map of individual flats, scammers use fake paper documents to take loans on the same flat from two different banks.
- **Parking and Terrace Disputes:** Builders often sell the same basement parking spot to two different buyers, leading to endless arguments and court cases.
- **Property Ownership Fraud:** Without a 3D digital record, proving exact boundaries and vertical heights of an apartment is difficult, leading to legal battles that drag on for years.

2. Our Solution: What Are We Building?

We are building a **3D Web Mapping System and 3D ULPIN Generator** that brings flat 2D land records into the 3D digital world.

Key Features of Our Solution:

1. **Unique 3D-ULPIN for Every Flat:**
   Our system automatically generates a unique code for every apartment unit by adding the vertical height (Z-axis) and floor number to the original ground ULPIN.
   - *Formula:* Base Land ULPIN + Floor Number + Unit Number + Height in Meters.
   - *Example Code:* `14MH2704291845-FL04-U402-Z12.5`
2. **Interactive 3D Building Explorer (Web App):**
   Anyone can open the website in their phone or laptop browser and see an interactive 3D model of an apartment building. You can rotate, zoom in, and click on any floor.
3. **Instant Property Verification:**
   Clicking on any flat turns it green and opens a side panel showing:
   - Flat owner's name
   - Carpet area in square feet
   - Assigned car parking spot
   - Loan / bank mortgage status

4. **Smart Fraud Blocker (Conflict Detection):**
   If someone tries to assign the same parking slot to two different flats, or register an illegal structure on the terrace, the system detects the overlap in 3D coordinates and immediately shows a **Red Warning Alert**.

3. How Is This Useful for the Government and People?

For Normal People and Homebuyers:

- **No More Buying Scams:** Before buying a flat, you can search its 3D ULPIN on the website and verify if the builder actually owns the unit and has legal approval.
- **Protected Parking Rights:** Your parking slot is digitally locked to your 3D ULPIN, stopping anyone else from claiming or selling your space.
- **Faster Home Loans:** Banks can verify property records in seconds instead of sending inspection agents for weeks.

For Banks:

- **Zero Double-Mortgage Fraud:** A bank can instantly see if another financial institution has already registered a loan against that exact 3D ULPIN.

For Municipal Corporations and the Government:

- **Accurate Property Tax Collection:** Municipalities can see all floors and illegal extensions in 3D, stopping tax evasion on unrecorded construction.
- **Fewer Court Cases:** Resolves boundary and common-area disputes before they reach the courts.
- **Foundation for Smart Cities:** Essential digital infrastructure for modern urban planning, underground metro lines, and utility pipelines.

4. Proper Tech Stack & Datasets (100% Free & Open-Source)

You do not need to spend any money or enter credit card details. The entire project runs on free tiers:

The Tech Stack:

Layer	Tool / Library	Cost	Why We Use It
3D Rendering	Three.js / React Three Fiber	100% Free	Renders smooth 3D buildings directly inside the user's web browser without requiring expensive cloud GPU servers.
Frontend UI	React.js / Next.js with Tailwind CSS	100% Free	Fast, modern web interface with responsive mobile and desktop support.
Database	Supabase (PostgreSQL + PostGIS)	100% Free Tier	Stores flat details, ownership data, and 3D spatial coordinates.
Backend API	Node.js / Python FastAPI	100% Free	Handles 3D-ULPIN generation and verification logic.
Hosting	Vercel	100% Free	Instant deployment providing a public `.vercel.app` link for the judges.

Datasets Needed & Where to Get Them for Free:

1. **Building Footprints & Geometries:**
    - Source: **OpenStreetMap (OSM)** via the **Overpass Turbo API**.
    - How to get it: Search for any area in your city (e.g., Nashik or Pune), export the building polygons as a `.geojson` file for free.
2. **Sample Land Records (2D ULPIN):**
    - Source: Public land portals like **Bhulekh / Mahabhulekh**.
    - Use sample publicly available survey numbers and parcel shapes as mock ground parcels.
3. **Apartment & Flat Floor Plans:**
    - Source: Public project filings on state RERA portals (e.g., **MahaRERA** public project disclosures) or create a standard template layout (e.g., 4 flats per floor with 2 bedrooms each).